ChatGPT

Simple Explanation Request

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Source: https://chatgpt.com/c/698c0f39-6154-832a-a169-a41e489f5d6d

For a large population, the probability of a random variable falls above $\mu + 2\,\sigma$ is:

- 0.015
- 0.035
- 0.025
- 0.25

explain properly in a simple language so i can understand the concept

Short answer: **About 0.0228 (≈ 0.023)**, so the best choice from the options is **0.025**.

Why — in plain simple steps:

1. A Normal (bell-shaped) distribution is symmetric around the mean μ.
   That means the probability of being more than 2σ above the mean equals the probability of being more than 2σ below the mean.
2. Standardize the value $x = \mu + 2\sigma$:
   $z = \dfrac{x - \mu}{\sigma} = \dfrac{\mu + 2\sigma - \mu}{\sigma} = 2.$
   So we want the area in the right tail for $z = 2$: $P(Z > 2)$.
3. Use the empirical rule (easy memory trick):
   - About **95.45%** of values lie within $\mu \pm 2\sigma$.
   - So about **4.55%** lie outside that interval (both tails combined).
   - Because of symmetry, half of 4.55% is in the upper tail: $4.55\%/2 = 2.275\%$.
   - As a decimal that is $0.02275 \approx 0.0228$.
4. A more exact look at the standard normal table gives $\Phi(2) \approx 0.97725$, so $1 - 0.97725 = 0.02275$ — same result.

Compare to the answer choices: 0.015, 0.035, **0.025**, 0.25 — **0.025** is the closest to 0.0228, so pick that.

Intuition: being more than two standard deviations above the mean is **rare** — only about 2 out of every 100 observations.

Use the empirical rule (easy memory trick): * About **95.45%** of values lie within μ±2σ\\mu\\pm2\\sigmaμ±2σ. * So about **4.55%** lie outside that interval (both tails combined). * Because of symmetry, half of 4.55% is in the upper tail: 4.55%/2\=2.275%4.55\\%/2 = 2.275\\%4.55%/2\=2.275%. * As a decimal that is 0.02275≈0.02280.02275 \\approx 0.02280.02275≈0.0228

explain more properly

Okay Vasu, let's go very slowly and clearly step by step.

We are finding:

$$P(X > \mu + 2\sigma)$$

This means:
"What is the probability that a value is more than 2 standard deviations above the mean?"

Step 1: Understand the Normal Distribution

A normal distribution:

- Is bell shaped
- Is symmetric (left side = right side)
- The mean is in the center

So whatever happens on the right side also happens on the left side.

Step 2: Use the Empirical Rule (Very Important for Exams)

You should remember this rule:

- About **68%** of data is within ±1σ
- About **95%** of data is within ±2σ
- About **99.7%** of data is within ±3σ

We are working with **±2σ**, so:

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 95\%$$

More accurate value: **95.45%**

Step 3: What is Outside ±2σ?

If 95.45% is inside, then the remaining must be outside.

Total probability = 100%

$$100\% - 95.45\% = 4.55\%$$

So:

- 4.55% is outside ±2σ
- That includes BOTH tails (left + right)

Step 4: Because of Symmetry

The normal distribution is symmetric.

So that 4.55% is split equally:

$$\frac{4.55\%}{2} = 2.275\%$$

So:

- Left tail = 2.275%
- Right tail = 2.275%

Step 5: Convert to Decimal

$$2.275\% = 0.02275$$

Approximately:

$$0.0228$$

Final Answer

$$P(X > \mu + 2\sigma) \approx 0.0228$$

From your options, the closest value is:

$$\boxed{0.025}$$

Simple Intuition

If 1000 students write an exam:

Only about **23 students** will score more than 2 standard deviations above the mean.

It is rare — but not extremely rare.

If you want, I can also explain using **Z-table method** step by step (good for exam preparation).

$\pm 1\sigma$
explain the sign what is it ?

Good question 👍 Vasu.

You are asking about this sign:

$$\pm 1\sigma$$

Let's understand it clearly.

What does ± mean?

The symbol **±** means:

> **plus or minus**

So,

$$\pm 1\sigma$$

means:

$$+1\sigma \quad \text{and} \quad -1\sigma$$

Both sides.

What is σ (sigma)?

- σ = standard deviation
- It tells us how far values are spread from the mean.

What does ±1σ mean?

It means:

$$\mu - 1\sigma \quad \text{to} \quad \mu + 1\sigma$$

So we go:

- 1 standard deviation **below** the mean
- 1 standard deviation **above** the mean

That gives us the interval around the mean.

Example (Simple Numbers)

Suppose:

Mean $\mu = 50$
Standard deviation $\sigma = 10$

Then:

$$\mu \pm 1\sigma$$

means:

$$50 - 10 = 40$$

$$50 + 10 = 60$$

So the interval is:

$$40 \text{ to } 60$$

That is ±1σ.

Why do we use ± ?

Because the normal distribution is symmetric.

So when we say:

- ±1σ → both sides
- ±2σ → both sides
- ±3σ → both sides

We are talking about a **range around the mean**.

Quick Summary

Symbol	Meaning
+1σ	1 standard deviation above mean
-1σ	1 standard deviation below mean
±1σ	Both +1σ and -1σ

If you want, I can also draw a simple diagram explanation in words so you can visualize it easily for exams.

95%
how ?

Ahh okay 👍 you are asking:

How do we know it is 95% inside ±2σ?

Very good question. Let's explain properly.

1️⃣ Where does 95% come from?

It comes from the **standard normal distribution table (Z-table)**.

When we convert everything to Z-scores:

$$Z = \frac{X - \mu}{\sigma}$$

For:

$$\mu + 2\sigma$$

we get:

$$Z = 2$$

For:

$$\mu - 2\sigma$$

we get:

$$Z = -2$$

So we want:

$$P(-2 < Z < 2)$$

2️⃣ Look at Z-table value for 2

From Z-table:

$$P(Z < 2) = 0.9772$$

Because of symmetry:

$$P(Z < -2) = 0.0228$$

3️⃣ Now subtract

$$P(-2 < Z < 2) = P(Z < 2) - P(Z < -2)$$

$$= 0.9772 - 0.0228$$

$$= 0.9544$$

That is:

$$95.44\%$$

We round it to:

$$\approx 95\%$$

4️⃣ Why teachers say 95%?

Because of the **Empirical Rule (68–95–99.7 rule)**:

- ±1σ → 68%
- ±2σ → 95%
- ±3σ → 99.7%

It is a simplified memory rule for exams.

The exact value is 95.45%.

5️⃣ Simple Intuition

If 1000 people are measured:

About **954 people** will fall within ±2σ.

Only about **46 people** fall outside.

And only **23 people** are above +2σ.

So the 95% is not random.

It comes from:

✔️ Z-table calculation
✔️ Area under normal curve
✔️ Symmetry

If you want, I can also explain **why the normal curve area becomes 0.9772 at Z=2** (conceptually, not just table).